Yufeng Huang

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Education

Huazhong University of Science and Technology **Sep 2014 – Jun 2017**

M.S. in Pattern Recognition and Intelligent Systems • GPA: 3.75/4.0 *Wuhan, China*

- Thesis: *CNN-based Visual Place Recognition in Changing Environments*
- Relevant coursework: Machine Learning and Data Mining, Pattern Recognition, Computer Vision, Theory of Matrices, Stochastic Processes, Digital Image Processing

Huazhong University of Science and Technology **Sep 2010 – Jun 2014**

B.E. in Measurement & Control Technology and Instrumentation • GPA: 3.73/4.0 *Wuhan, China*

- Relevant coursework: C Programming, Data Structures, Computer Networks, Calculus, Linear Algebra, Probability Theory, Digital Signal Processing

Professional Experience

Tencent — SGRA Studio **Jul 2024 – Present**

Tools Engineering Team Lead & Game AI Lead *Guangzhou, China*

Role Overview: Lead SGRA Studio's tools engineering team and serve as the studio's Game AI lead, overseeing development tools, AI-assisted production workflows, and game AI initiatives.

- **LLM-driven NPC agents**: Designed and built a hybrid NPC AI architecture for *Animula Nook*, combining behavior trees, finite-state machines, utility AI, and LLM-based modules. Each NPC exhibits configurable personality, maintains conversation memory, performs autonomous daily routines, and engages in personalized interaction with players. Per scene, dozens of fully autonomous NPCs operate concurrently; at server scale, the architecture supports millions of NPC entities (10x player population) via behavioral level-of-detail (LOD) and resource scheduling.
- **AI-assisted co-creation tools**: Led development of 10+ LLM-powered authoring tools spanning the full content production pipeline, including narrative and dialogue generation, behavior-tree generation from natural language, level generation, and automated cutscene generation (converting narrative scripts into full cinematics with character animation, expression, and camera choreography). These tools substantially reduced content-creation time and directly explored human-AI co-creation workflows at production scale.
- **Agent-driven tool and game generation**: Built an experimental pipeline in which an LLM agent autonomously generates production tools, from UI design through code implementation and automated testing. Also exploring multi-agent approaches to general-purpose game generation to accelerate playable game demo development and investigate more autonomous creation workflows.
- **Gameplay systems**: Contributed to core gameplay systems including character animation, NPC systems, dialogue, quests, combat, and in-game chat.
- **Infrastructure**: Designed and built the globally distributed server architecture for *Animula Nook*, supporting 1M+ DAU in a single shared world.

ByteDance — Nuverse **Jun 2021 – Dec 2023 & Feb 2024 – Jul 2024**

Senior Software Engineer *Guangzhou, China*

Role Overview: Developed game AI systems and production tools across multiple studio projects, spanning tactical combat AI, open-world ambient NPC behavior, and generative AI for content.

- **Auto-chess tactical AI**: Built the battle AI for *Dragonheir: Silent Gods* (launched strategy RPG with millions of downloads) using behavior trees and state machines for auto-chess-style tactical combat. Explored reinforcement learning for hero lineup composition and positioning strategies.
- **Souls-like boss AI**: Designed complex multi-phase boss AI (comparable to Souls-like titles) using hierarchical behavior trees and state machines, modeling player-adaptive attack patterns and phase transitions.
- **Narrative NPC AI**: Built an event-driven, blueprint-based NPC AI system for *Dragonheir: Silent Gods* supporting 1,000+ concurrent NPCs per scene, powering extensive scripted narrative performances in a text-heavy D&D-themed CRPG.

- **Open-world NPC AI**: Developed a utility-based ambient NPC behavior system for an internal open-world ARPG, enabling context-aware daily routines and environmental responses.
- **AIGC for games**: Explored generative AI in game production, including concept art and UI icon generation with diffusion models, text-to-motion generation, and video-based motion capture.
- **Production / authoring tools**: Developed a suite of internal tools, including but not limited to blueprint editors, behavior-tree editors, narrative editors, quest editors, and dialogue editors.

NetEase Games **Jul 2017 – Jul 2019 & Apr 2020 – Jun 2021**
Software Engineer *Guangzhou, China*

Role Overview: Developed gameplay and AI systems for open-world MMORPG and MOBA projects.

- Developed gameplay systems (character control, camera, level system) for an internal open-world MMORPG project.
- Built companion (buddy) AI combining behavior trees and utility AI, enabling NPC allies to fight alongside players with context-aware combat behavior.
- Gained working knowledge of utility-based MOBA combat AI on *Onmyoji Arena*, and participated in exploratory research on reinforcement-learning-based PvP agents.
- Shipped multiple gameplay systems for *Onmyoji Arena* (launched MOBA title), including hero skills, in-game activity systems, and battle pass systems.

Publications

1. **Y. Huang** and J. Liu. "CNN-based visual place recognition method." *Computer & Digital Engineering*, 2018(6). Link
2. **Y. Huang**, Y. Liu, and J. Liu. "New bionic navigation algorithm based on the visual navigation mechanism of bees." *Proceedings of SPIE*, 9522, pp. 235–240, 2015. DOI
3. L. Wang, **Y. Huang**, and J. Liu. "Bionic navigation algorithm based on contrast entropy flow." *China Sciencepaper*, 2015(10), pp. 967–970. Link

Honors & Awards

- **Second-Class Graduate Academic Scholarship**, Huazhong University of Science and Technology, 2015–2016 and 2016–2017
- **Meritorious Winner**, Mathematical Contest in Modeling (MCM), COMAP, 2013

Technical Skills

Programming: Python, C/C++, C#
Software and Infrastructure: FastAPI, Docker, Linux, distributed server architecture
Game AI: Behavior trees, finite-state machines, utility AI, RL, LLM-driven agents
AI Frameworks: PyTorch, Hugging Face Transformers, LangChain
Game Engines: Unity, Unreal Engine, Godot